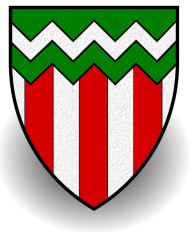
Curriculum Vitae of Cameron KERR



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Employment History

August 2023 – current **Senior Integrations Engineer**
Tabula (prev. TracMap NZ) Permanent Full-Time On-site

- As a team of two, I am tasked with analysing and delivering frontend and backend changes to Tabula Online, an agritech SaaS platform with a strong GIS flavour. This involves work throughout the frontend and backend and L3 support for our customers and B2B API partners.
- **Proved myself as capable in maintaining large existing code-bases in Javascript, Python and PostgreSQL with minimal guidance.**
- Routinely practise and benefit from standard development practices around code contribution, testing and review.
- Analyse and design integration solutions for various partners.
- Used AI assisted tools (namely Claude Code) for the bulk of my code since 2025. Focus on retaining critical thinking, owning the code generated, balancing risk, delivering value quickly, running experiments that would otherwise be infeasible and paying down technical debt.

May 2022 – August 2023 **Site Reliability Consultant**
Pythian Services, Inc., Canada. Permanent Full-Time 100% Remote

- Part of a respected global Managed Services Provider in a Linux SRE team helping clients in multiple time zones with their in-cloud and on-premises workloads.
- Gained practical experience with Amazon and Azure to go along with my recent AWS Solutions Architect Associate (and later AWS SysOps Associate).
- Worked with clients in various industries in a commercial environment, including banking, health, cyber-security and higher-education.
- Helped clients manage their Open Source software estate within environments such as on-premises, VMs in the cloud, containers and serverless.
- Recognised for my skills in conscientiously documenting my work and creating training materials for clients and peers.

May 2019 – May 2022 **Promoted to Group Lead**
University of Otago, Dunedin, New Zealand. Permanent Full-Time.
Further restructuring opened the door to continuous improvement / research-and-development and a dedicated cross-functional troubleshooting resource; essentially an SRE. Still working in the IT Infrastructure (Systems Services) team inside Information Technology Services, University of Otago.

May 2011 – May 2019 **(Senior) Systems Engineer**
University of Otago, Dunedin, New Zealand. Permanent Full-Time.
Working in the Systems Services team inside Information Technology Services, University of Otago, my tasks were largely relating to specialisations in Linux, web infrastructure and corporate systems. Restructuring near the end created a new Senior position, into which I was promoted.

Feb. 2002 – May 2011 **Teaching Fellow**
University of Otago, Dunedin, New Zealand. Permanent Full-Time.
Working in the Computer Science Department. Duties included design, implementation and delivery of teaching materials including laboratory manuals, assignments and some lectures; significant programming, systems integration and administration to support these papers. This role requires a large amount of self-management regarding course design, classroom management, time and workload.
Mixed full- and part-time while completing PgDipSci in Computer Science.

Feb. 2001 – Feb. 2002 **Research Fellow**
University of Otago, Dunedin, New Zealand. Fixed-term.
Working under Dr. Zhiyi Huang, Computer Science Department.
Duties included research into distributed computing. Some light teaching responsibilities.

Summer of 2001/2 **Summer Bursary**, University of Otago, Dunedin, New Zealand
Working under Dr. Zhiyi Huang, Computer Science Department.
Duties included research into distributed computing and cluster computing

Late 1998 – Early 2000 **IT Support Technician** Willis White & Company Ltd, New Zealand
Contracted to work in the Dunedin City Council
Duties included rollout of Windows 95 workstations, Y2K preparation and general IT Support

Certifications

AWS Certified Solutions Architect - Associate
Expires: Mar 27, 2025
Validation Number: FZTBQ73DFFF11GC0 (Validate: https://aws.amazon.com/verification)

AWS Certified SysOps Administrator - Associate
Expires: Jan 03, 2026
Validation Number: PDGQYRZKQNF11K3P (Validate: https://aws.amazon.com/verification)

Education

Postgraduate Diploma of Science in Computer Science (with Distinction)
University of Otago, New Zealand, 2009

Bachelor of Science Degree in Computer Science
University of Otago, New Zealand, 2001

Recent Experiences

Here are some recent highlights, showing the types of work I've felt most proud of and have enjoyed. They demonstrate insight and innovation to solve problems, particularly surrounding Open Source Software:

- A key business partner was struggling with API performance issues with a new, larger type of workload. I undertook some exploratory work with Claude Code to iterate re-architecting a critical part of our product, involving PostGIS, Python, C++, algorithmic and database optimisations. This gave us a set of very significant improvements we could implement in future according to risk appetite and downstream testing.

- A new business partner was looking to integrate GIS data, and I was asked to quickly create a design proposal and then to begin prototype development using the materials provided. Using Claude Code I was able to quickly iterate to a polished demonstration inside of a week (original estimate 6 weeks) with relatively few open questions ahead of a meeting with the partner. AI-assisted tools have changed the calculus of prototyping and integration, and freed me to think more about data-modelling and UX design.

- A Higher Education client had fundamental performance issues with its LDAP service, which was denying users on its campus access to the wireless network. Actually useful performance metrics were hard to obtain in the version of the software encountered but with some deeper investigation the performance patterns could be visualised and understood; immediate changes could be made to make access patterns more efficient and further architectural recommendations were provided for consideration.

- A Media & Communications business unit had implemented a live-streaming video service that had aged and was failing to scale beyond about 140 concurrent users. This was causing highly visible outages and stress with the most senior levels of management. I was brought in to understand the root causes of the issue and remediate. After analysis, I researched and deployed a streaming media caching layer using Nginx, and load-tested it by customising a third-party research project for benchmarking HTTP streaming. It could then support about 2.5k concurrent users

per proxy, with much improved visibility and monitoring, and a key event of the year went off without a hitch.

- A large-scale custom data platform for COVID-19 Contact Tracing, based on connection data from one of the largest Wi-Fi deployments in the southern hemisphere, had developed business critical performance issues during reporting after a previous developer had added necessary privacy enhancements. I analysed the entire system, which was based on Python and TimescaleDB (PostgreSQL with time-series extensions), re-architected how encryption was being used to maintain security and privacy while radically improving query-time performance (from over a day to about ten minutes). I performed the necessary data-migration on the live production system, and at the same time greatly streamlined the reporting workflow with some user-friendly tooling and using Jupyter Notebooks, Pandas, etc. to create reports.

- I've managed the Web Infrastructure for an Enterprise-scale off-the-shelf CMS from Oracle, which had many architectural failings due to the product design, and licensing and the business requirements it was meant to fulfill. To make it as robust as possible, I implemented in-depth observability using the Elasticsearch, Logstash and Kibana (ELK) stack, leading to a series of data-driven optimisations around reverse-proxying, caching, WebLogic middleware optimisation and working with developers on changing how data-integrations could be rearchitected.

- I've designed, deployed and maintained a high volume, clustered data-streaming service based on Red Hat Clustering, Kafka, Logstash, Elasticsearch, Prometheus, Grafana, Docker, with numerous critical bespoke integrations in Python, Go, Ruby and more, and completely deployed and maintained using Ansible.